Concepts and Models

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Outline

Key Components

Cloud Service Models

Conceptual Reference Architecture

Cloud Deployment Models

Key Components – NIST 2011 I

Five Essential Characteristics

Four Deployment Models

Three Service Models

Key Components – NIST 2011 II

Essential Characteristics

1. On-demand Self-service – via a service portal
2. Broad Network Access – ubiquitous access
3. Resource Pooling – location independent, multi-tenancy ⇐
4. Rapid Elasticity – time to market/fast deployment
5. Measured Service – pay-per-use

Key Components – NIST 2011 III

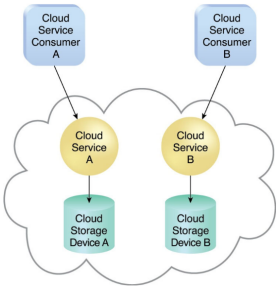
Deployment Models

1. Public Cloud
2. Private Cloud
3. Community Cloud
4. Hybrid Cloud

Service Models

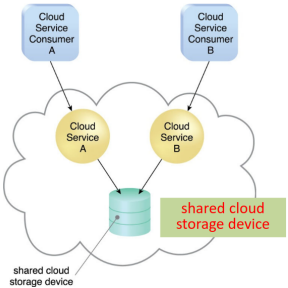
1. Software-as-a-Service (SaaS)
2. Platform-as-a-Service (PaaS)
3. Infrastructure-as-a-Service (IaaS)

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Single Tenant

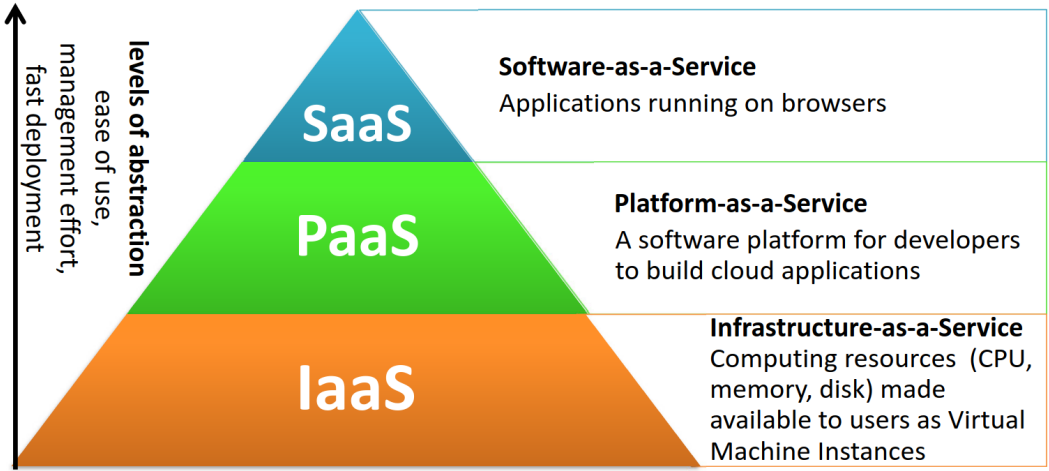
- Each cloud consumer has a **separate** IT resource instance



Multi-Tenancy

- Cloud provider pools its IT resources to serve multiple cloud service consumers
- **An instance** of the program serves multiple cloud consumers (tenants)

Cloud Service Models



Cloud Service Models

Software-as-a-Service (SaaS)

Use provider's applications over a network
E.g.: Salesforce.com, Google apps, Microsoft 365, ...

Platform-as-a-Service (PaaS)

Deploy customer-related applications to a cloud
E.g.: Google's app engine, Heroku, Microsoft Azure App Service, ...

Infrastructure-as-a-Service (IaaS)

Rent processing, storage, network capacity, etc.
E.g.: Amazon EC2, DigitalOcean, Linode, Alibaba ECS, ...

Cloud Service Models

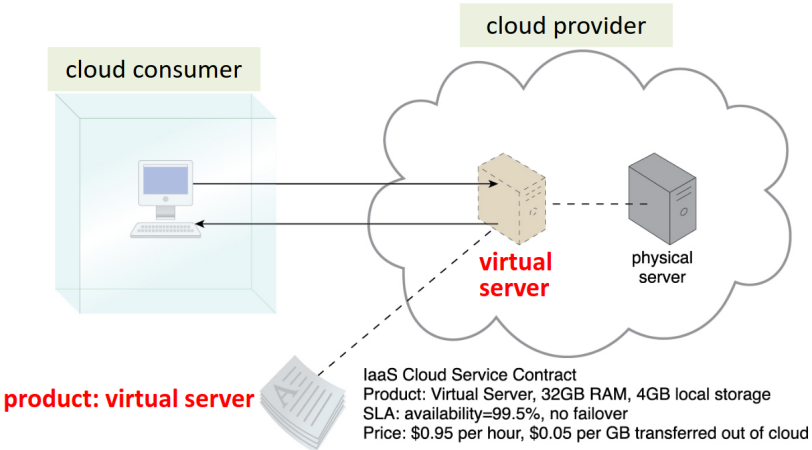
Why different models?

Different levels of abstraction and management – use depending on need!

Infrastructure-as-a-Service (IaaS)

Lowest level of abstraction

Physical resources are abstracted as virtual resources



Servers

Storage

Networks

Applications & Middleware

...

Platform-as-a-Service (PaaS)

Two parts: Platform Software, and Computing Resources needed to run the platform software

PaaS Provider

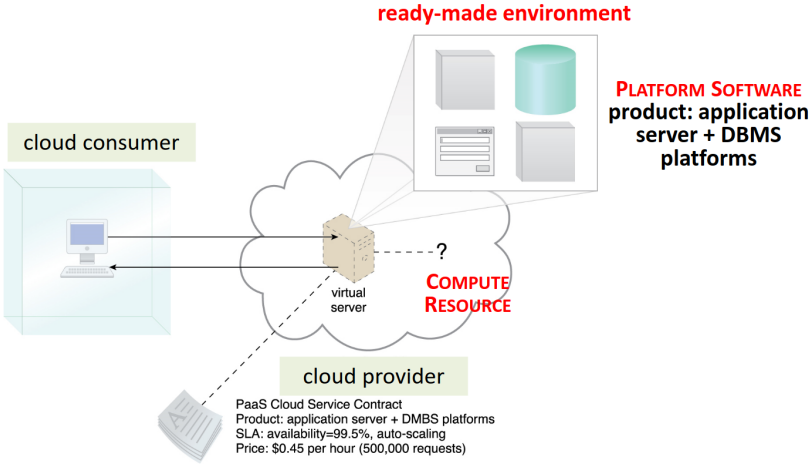
Develops the platform software and decides where to host/run the PaaS

PaaS Consumer

Incorporates the ready-made PaaS into the application being developed

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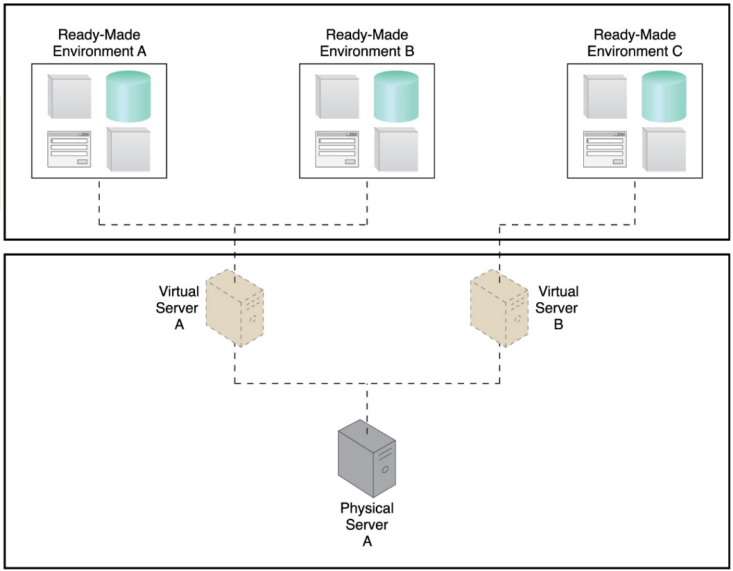
Platform-as-a-Service (PaaS)



Combining IaaS and PaaS

PaaS platform software

IaaS compute resource runs platform software



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Example: Combining IaaS and PaaS

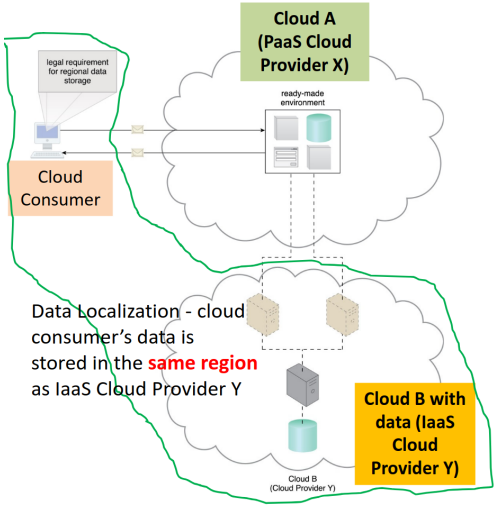
Meeting Legal Requirements for Data Storage

Cloud consumer subscribes to PaaS offered by provider **X (Cloud A)**

To meet requirement

Services offered by provider **X** are physically hosted on provider **Y (Cloud B)**

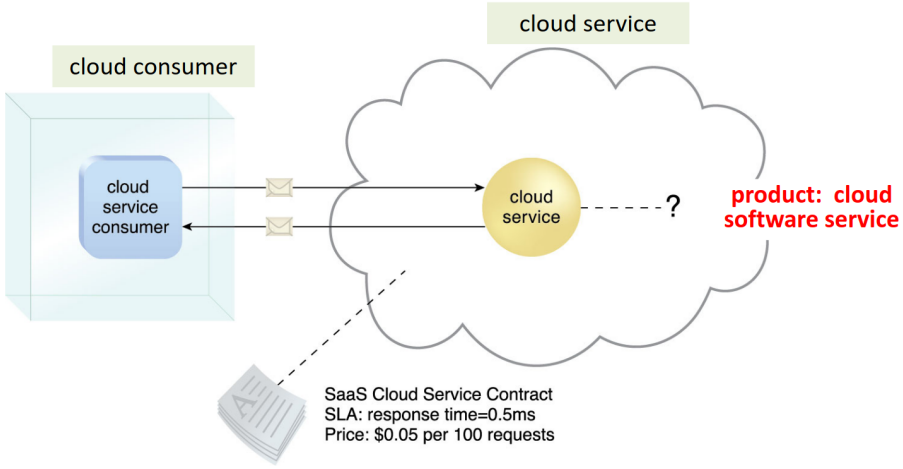
⇒ **Cloud B** is physically located in the same region as the cloud consumer



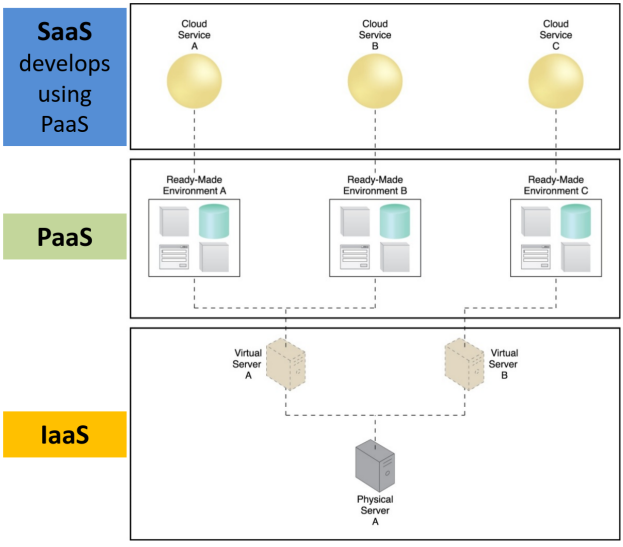
Software-as-a-Service (SaaS)

SaaS provider – installs and maintains software

SaaS consumer – remotely runs the software on the cloud



Combining IaaS, PaaS, and SaaS



Variants of Service Models

Storage-as-a-Service

Database-as-a-Service

Security-as-a-Service

Communication-as-a-Service

Integration-as-a-Service

Testing–as-a-Service

Function-as-a-Service – serverless computing

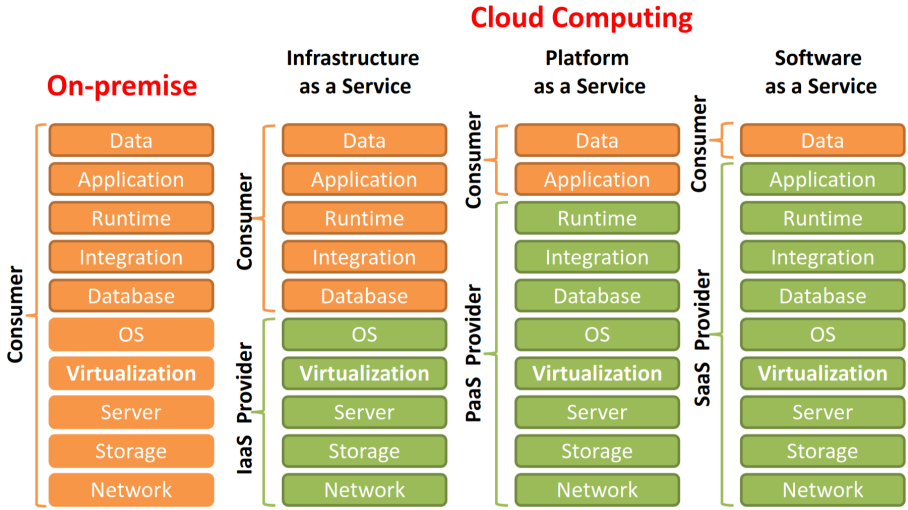
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Everything-as-a-Service

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Comparing Service Models

Responsibility of Consumer vs Provider



Comparing Service Models

Pros and Cons

Service Models	Pros	Cons
On-premise	Highest flexibility	**Long** time-to-market
IaaS	Scalability, No H/W procurement	Privacy
PaaS	DB, Frameworks, Middleware ready	Vendor lock-in
SaaS	**Reduce** time-to-market	Lowest flexibility

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Reference Model – NIST 2011

Focuses on the what (cloud services provide) and not the how (design solution and implementation)

Does not represent the system architecture of a specific cloud

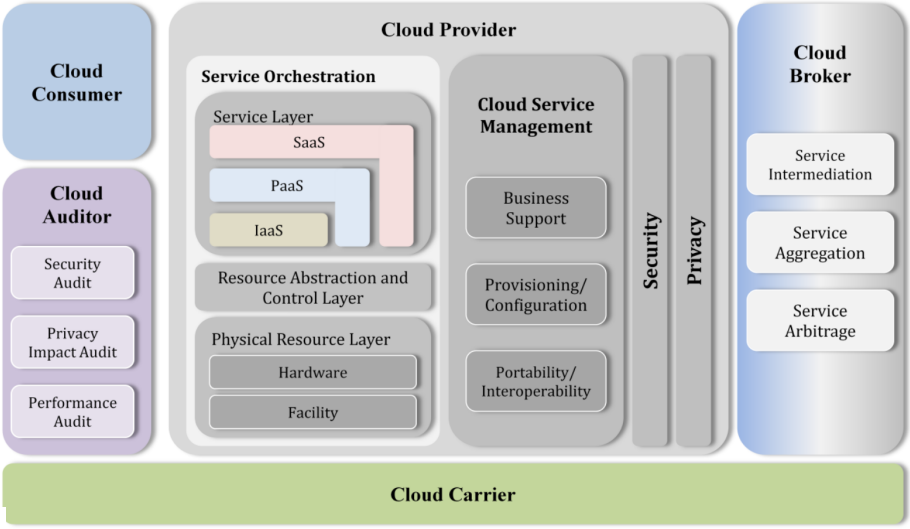
Provides a common reference framework to describe, discuss and develop a system-specific architecture

Vendor and Implementation Independent

https://doi.org/10.6028/NIST.SP.500-292

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Cloud Computing Reference Architecture



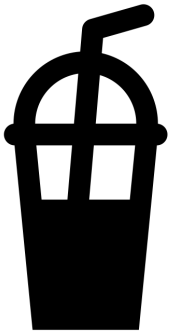
Actor

Entity (person/organisation) that participates in a transaction or process and/or performs tasks in cloud computing

Five major actors

1. **Cloud Consumer**
2. **Cloud Provider**
3. **Cloud Auditor**
4. **Cloud Broker**
5. **Cloud Carrier**

Take a Break







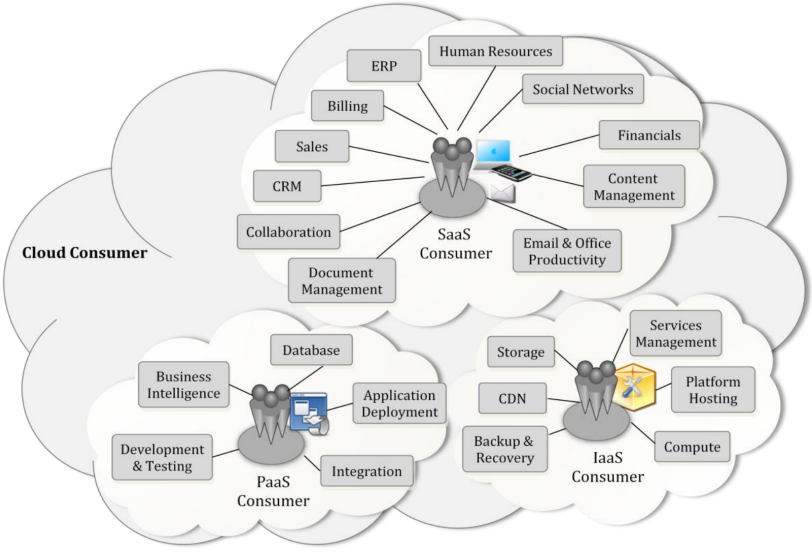
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Actor Roles

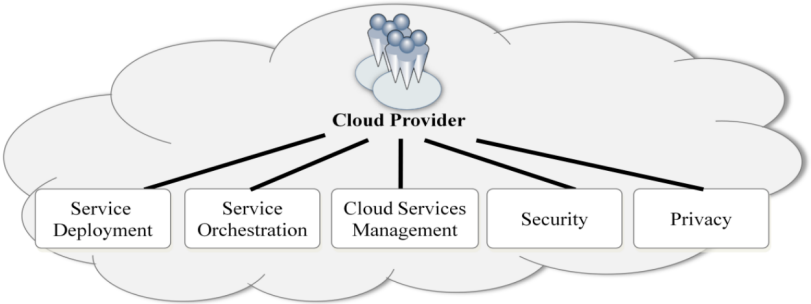
1. **Cloud Consumer** – maintains a business relationship with, and uses services from, **Cloud Providers**
2. **Cloud Provider** – offers a cloud service to **Cloud Consumers**
3. **Cloud Auditor** – conducts independent assessment of cloud services, system operations, performance and security of the cloud implementation
4. **Cloud Broker** – manages the use, performance and delivery of cloud services, and negotiates the relationships between **Cloud Providers** and **Cloud Consumers**
5. **Cloud Carrier** – provides connectivity and transport of cloud services from **Cloud Providers** to **Cloud Consumers**

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Cloud Consumer



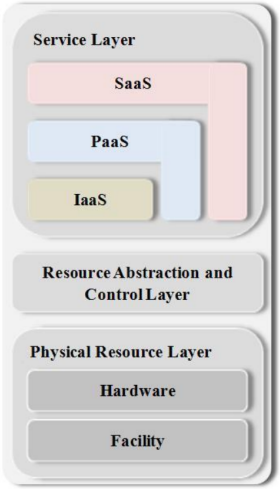
Cloud Provider



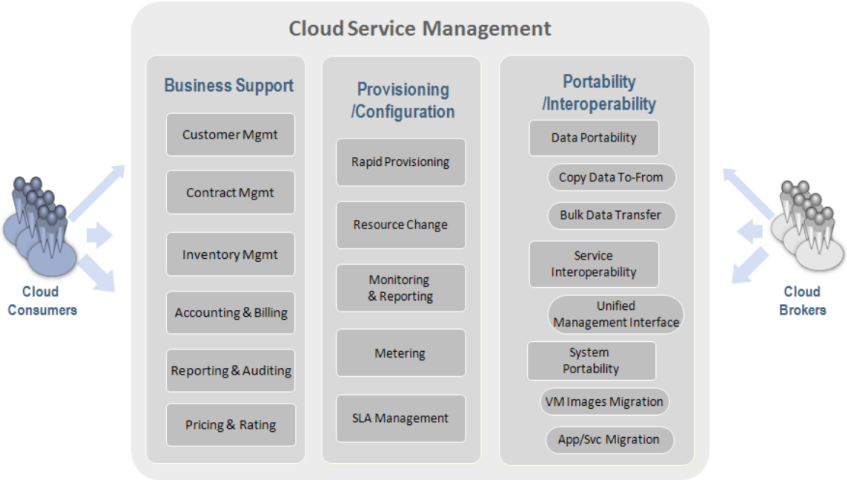
Depending on the service models, three types of cloud providers

1. SaaS cloud provider
2. PaaS cloud provider
3. IaaS cloud provider

Compose system components to support the cloud providers activities in arrangement, coordination and management of computing resources to provide cloud services to cloud consumers



Cloud Services Management



Service-related functions for the **management and operation** of the services provided to the cloud consumers

Cloud Service Management

Business Support – set of business-related services to deal with clients (consumers) and supporting processes such as accounting & billing, pricing & rating, ...

Provisioning/Configuration

- Rapid provisioning – automatically deploying cloud systems based on the requested services/resources/capabilities
- Resource change – adjusting configuration/resource assignment for repairs, upgrades and joining new nodes into the cloud
- ...

Portability/Interoperability

- Should provide mechanisms to support data portability, service interoperability, and system portability

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Security and Privacy

Security

- Cross-cutting and spans across all layers of the reference model
- Concerns all relevant actors!

Privacy

For Cloud Providers

- Personal Information (PI)
- Personally Identifiable Information (PII)

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Trust Boundary

Logical perimeter that represents which IT resources are trusted by an organisation

Implications of Cloud

- Increased Security Vulnerabilities
- Reduced Operational Governance Control
- Multi-Tenancy: Overlapping Trust Boundaries

Cloud Auditor

Independent examination of cloud service controls

- Security Controls
- Privacy Impact
- Performance
- ...

Verify compliance with regulation and security policy, and conformance to standards through review of objective evidence

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Cloud Broker

To help cloud consumers manage integration of cloud services

Three Main Services

1. Service Intermediation – provides value-added services to cloud consumers
2. Service Aggregation
   - Combines and integrates multiple services into one or more new services
   - Provides data integration
   - Ensures secure data movement between cloud consumers and multiple cloud providers
3. Service Arbitrage – flexibility to choose services from multiple agencies

Cloud Carrier

Provides connectivity and transport of cloud services between cloud consumers and providers

Cloud providers set up SLAs with a cloud carrier to provide consistent services to their consumers

May be required to provide dedicated and secured connections between cloud providers and consumers

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Cloud Deployment Models

Public Cloud

Available to the general public, i.e., **shared** by all consumers

Private Cloud

Solely for use by an organisation – suitable for enterprises with large scale IT

Community Cloud

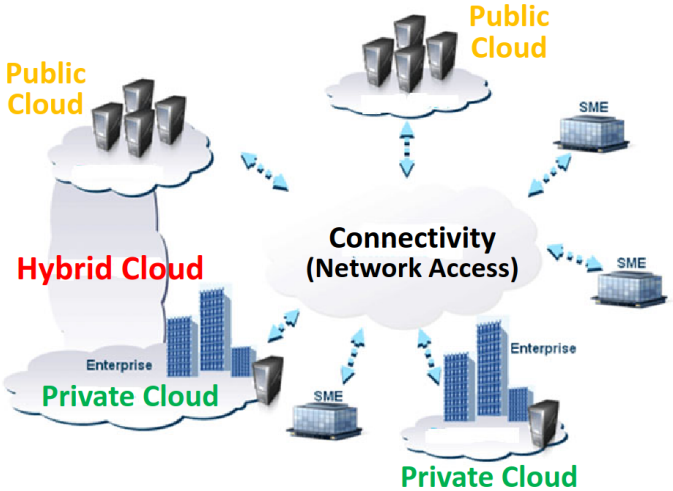
Shared by multiple organisations based on common operational and regulatory requirements

Hybrid (Federated) Cloud

Two or more public and private clouds that inter operate

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Cloud Deployment Models



Public Cloud

Most commonly used cloud infrastructure

Organisations act as cloud consumers while accessing IT resources and services made available by different cloud providers

Organisational Needs

- Low upfront costs with potentially a pay-as-you-go model
- Available anytime, anywhere
- Maintenance is handled by the provider

Cloud Providers

Amazon, Microsoft, Google, Oracle, IBM, Alibaba, Zoho, DigitalOcean, ...

Private Cloud

Cloud consumer's organisation has exclusive access and usage of cloud resources

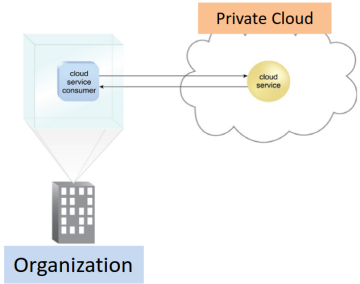
Organisational Needs

- High level of control – especially for sensitive data
- Restricted access – often limited to organisation
- Maintenance is handled predominantly in-house

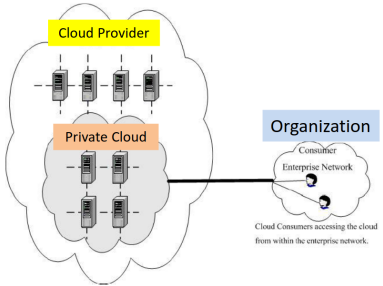
Can be managed on-site or by a third-party (out-sourced private cloud)

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Private Cloud



On-Site Private Cloud



Out-Sourced Private Cloud

Security? On-site vs out-sourced

Dedicated IT resources hosted by a third-party provider

Virtual Private Cloud – secure, isolated environment within a public cloud that is customisable by you

Private Cloud

Concerns

Limited SLA, Reliability, Security, Availability, Trust, ...

Cloud Providers

Amazon Virtual Private Cloud

Cisco Private Cloud Solutions

Virtual Private Cloud – Google Cloud

Azure Dedicated Host – Private Cloud

...

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Community Cloud

Positioned between public and private clouds

Organisational Needs

- Cost of private cloud is prohibitive
- Shared needs and regulatory compliance with other organisations
- Restricted access – often limited to participating organisations

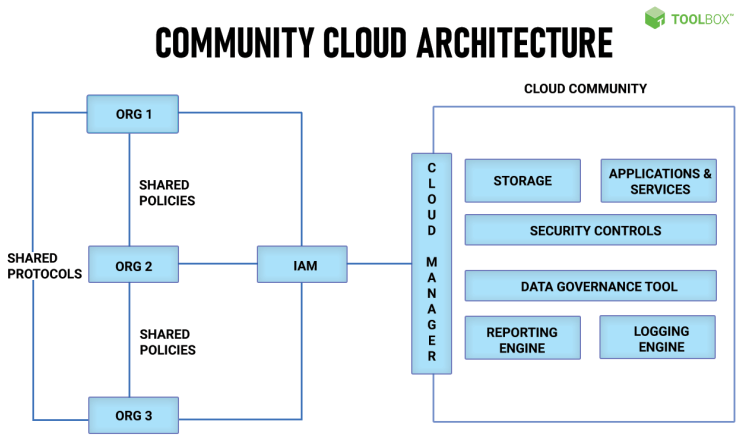
Need a clear governance structure for management

Examples

Government departments, Educational Institutions, Health Care Industry, ...

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Community Cloud



https://www.spiceworks.com/tech/cloud/articles/what-is-community-cloud/

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Hybrid Cloud

Consists of two or more clouds (on-site private, on-site community, off-site private off-site community or public) that remain as distinct entities

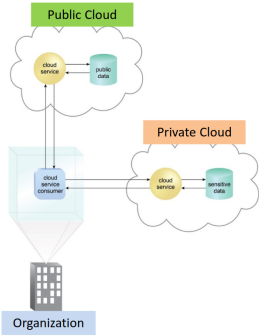
Connected together to allow for data and application interoperability

Organisational Needs

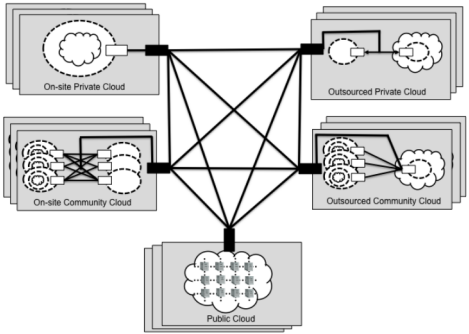
- Balance control and cost
- Flexible approach depending on data and applications, e.g., sensitive data on private cloud while application on public cloud

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Hybrid Cloud



2 different clouds



5 different clouds

Sovereign Cloud

Why?

- Meet regulatory and compliance needs of a country or jurisdiction
- Stores data (including metadata) within national borders

Organisational Needs

- Adhere to local data protection laws
- Prevents foreign access to data under all circumstances

Cloud Providers

Microsoft Sovereign Cloud, Sovereign Cloud Oracle, SAP Sovereign Cloud, ...

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Next Week

Cloud Architectures

References

NIST Cloud Computing Reference Architecture, NIST Report 2011, https://doi.org/10.6028/NIST.SP.500-292

The NIST Definition of Cloud Computing, NIST, 2011, https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf

Chapter 4, Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Prentice-Hall, 2013

Continuous Feedback



https://forms.cloud.microsoft/r/ZJxTf2wFjr

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